



GTC4Lusso USA - Shock absorbers - Latest Update: 2021-01-26

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D4.04 Shock absorbers

Removing the front shock absorbers

➤ Remove the engine compartment cosmetic shields ( E3.14).

 Remove lateral cosmetic shields.

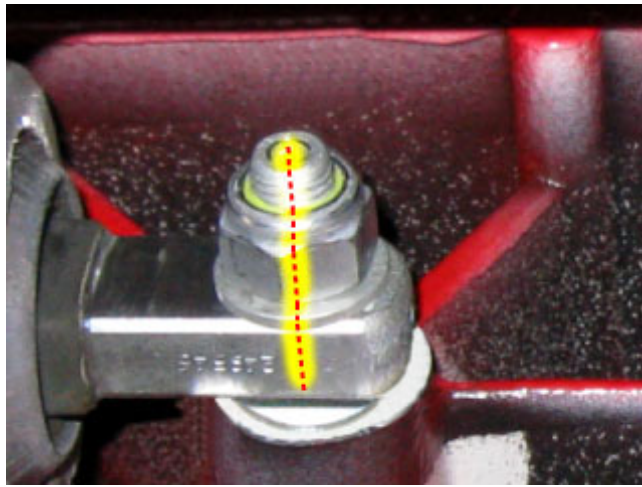
➤ Disconnect the battery ( F2.01).

➤ Remove the complete wheels ( D2.01).

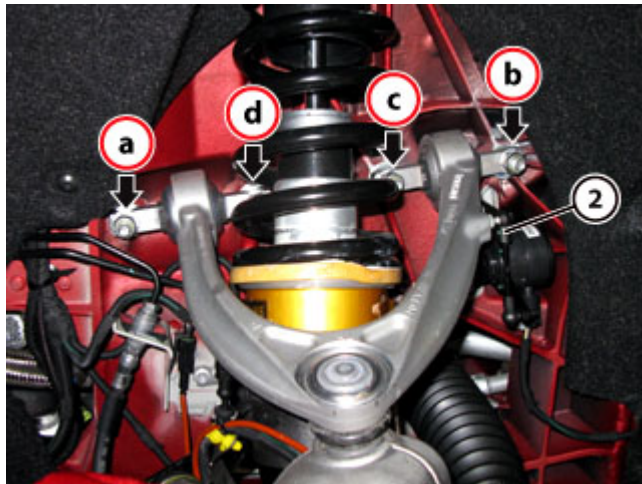
 Remove the front wheels.



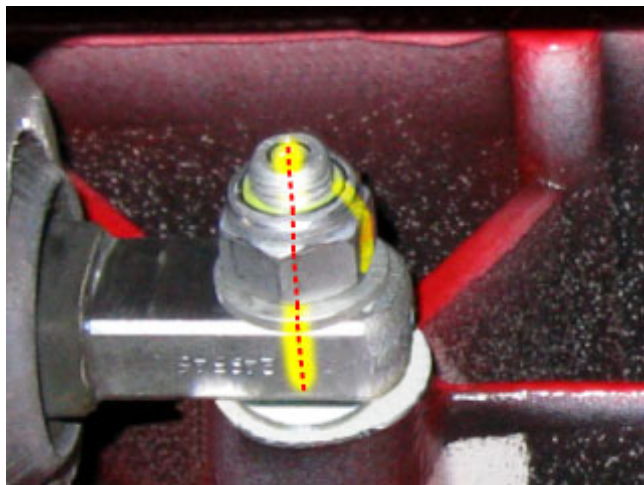
- Disconnect the connector (1).
- Detach the indicated clip.
- Undo the screws indicated.



- Using a delible yellow marker pen or paint, make reference markings on the stud bolts and relative nuts and the eyebolts of the Flanblocs fastening the upper arm.

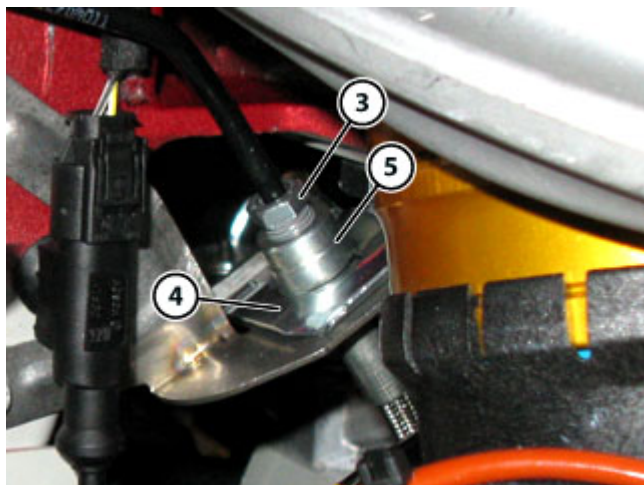


- Detach the tie-rod (2).
-  DO NOT use lubricants or electric/pneumatic drivers.
- Partially loosen the indicated nuts in the following sequence: (a), (b), (c), (d).
-  Move the shock absorber aside to facilitate the operation.



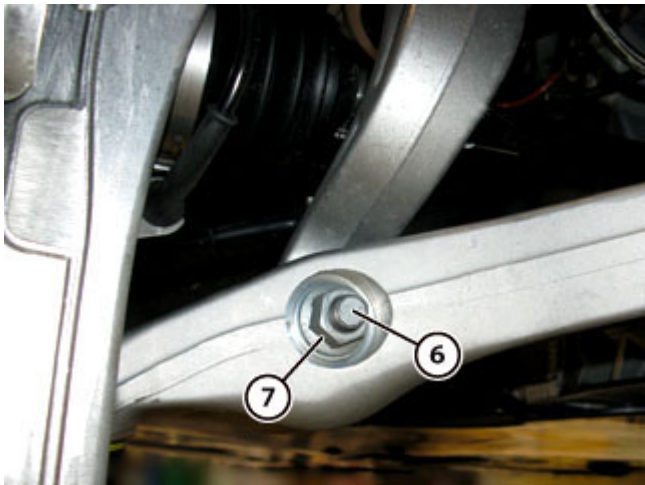
- Check that the yellow reference markings are correctly aligned: the marking on the stud bolts and the marking on the Flanbloc eyebolts must be aligned.
- If the markings on the stud bolts and eyebolts are not correctly aligned after loosening the arm fastener nuts (even if only on one stud bolt), a ROL (Red On Line) must be opened, attaching adequate photographic documentation, and contact the Ferrari Help Desk.
- If the markings on the stud bolts and eyebolts are correctly aligned after loosening the arm fastener nuts, completely undo and remove the four arm fastener nuts in the order given for the previous operation.
-  Move the shock absorber aside to facilitate the operation.

- Check again that the yellow reference markings are correctly aligned: the markings on the stud bolts and the markings on the Flanbloc eyebolts must be aligned.



For vehicles with suspension "Lift" system

- Undo and disconnect the union (3).
- Remove the clip (4).
- Detach the union (5).



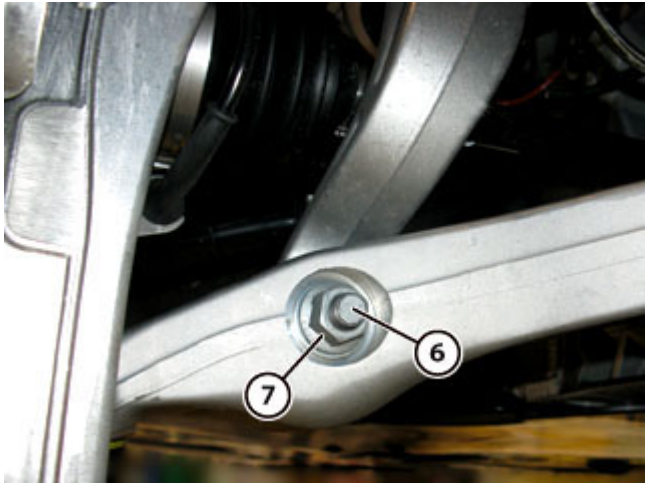
- Undo the nut (7), while preventing the pin (6) from turning.
- Remove the pin (6).

- Detach the upper arm from the relative stud bolts, turning slightly towards the front of the vehicle.
- Remove the shock absorber from the relative seat.

Refitting the front shock absorbers

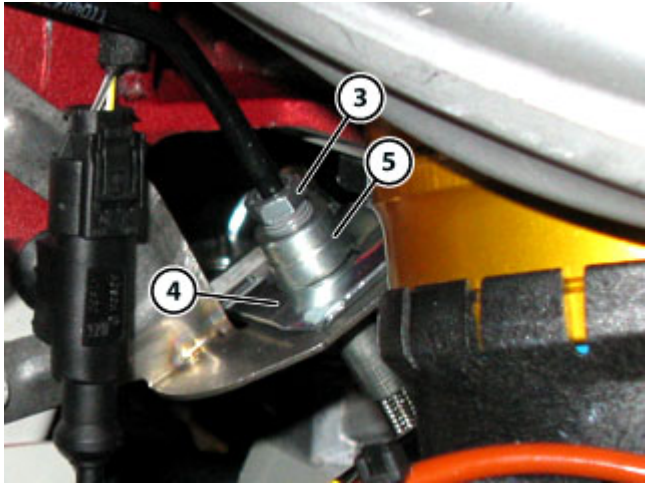
			
Tightening torque		Nm	Class
Fastening the front shock absorbers	Nut	80 Nm	A
	Screw (pretightening)	20 Nm	A
	Screw	110 Nm	A
	Union	16 Nm	B
Fastening the front suspension upper arms	Nut (pretightening)	52 Nm	A
	Nut	52 Nm	A
	Nut (pretightening)	9 Nm	A
	Nut	60 Nm	A

- Support the weight of the complete suspension adequately.
- Fit the shock absorber in the relevant seat.



- Install the pin (6).
- Tighten the nut (7), while preventing the pin (6) from turning.

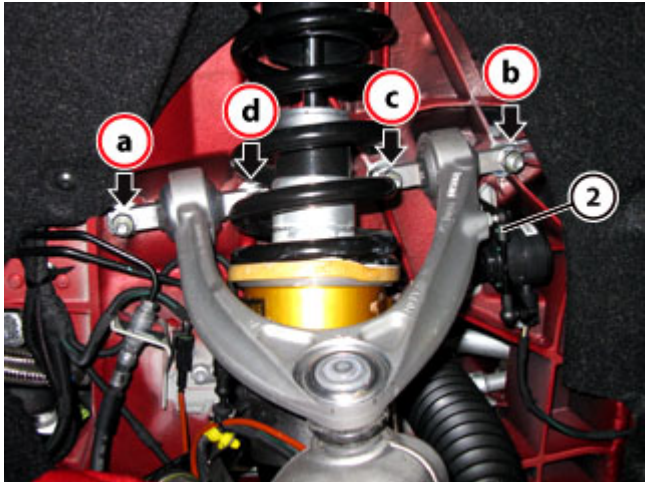
		
Tightening torque	Nm	Class
Nut	80 Nm	A



For vehicles with suspension "Lift" system

- Fit the union (5).
- Fit the clip (4).
- Connect and tighten the union (3).

		
Tightening torque	Nm	Class
Union	16 Nm	B



- Clean the mating surfaces with a degreasing product (heptane).
- Fit the upper arm on the relative stud bolts.
-  DO NOT use lubricants or electric/pneumatic drivers.
-  ALWAYS replace the suspension arm fastener nuts, complete with relative washers, each time they are removed. Replacement is NOT necessary if the nuts are loosened only.
- Tighten the indicated nuts in the following sequence: (a), (b), (c), (d).
-  Move the shock absorber aside to facilitate the operation.

		
Tightening torque	Nm	Class
Nut (pretightening)	9 Nm	A
Nut	60 Nm	A

- Attach the tie-rod (2).



- Check that the yellow reference markings are correctly aligned: the markings on the stud bolts and the markings on the Flanbloc eyebolts must be aligned.



- Remove the yellow reference markings made during removal.
- Using an indelible red marker pen or paint, make reference markings on the stud bolts and relative nuts and the eyebolts of the Flanblocs fastening the upper arm.



- Tighten the screws indicated.

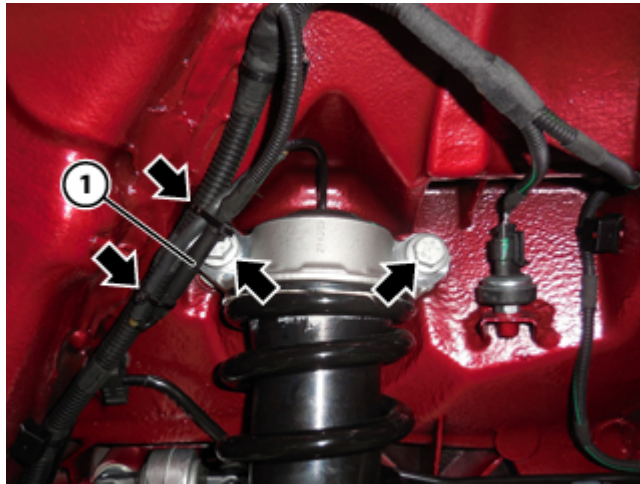
		
Tightening torque	Nm	Class
Screw (pretightening)	20 Nm	A
Screw	110 Nm	A

- Connect the connector (1) to the shock absorber.
- Fasten the indicated clip.

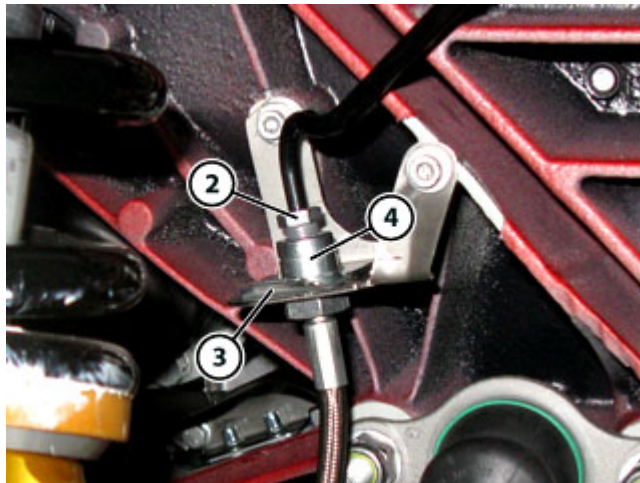
- Connect the battery ( F2.01).
- For vehicles with suspension "Lift" system. Fill and bleed the suspension "Lift" system ( D4.08).
- Refit the complete wheels ( D2.01).
- Refit the front wheels.
- Refit the engine compartment cosmetic shields ( E3.14).
- Refit the lateral cosmetic shields.
- Adjust the vehicle suspension/wheel alignment ( D4.06).

Removing the rear shock absorbers

- Disconnect the battery ( F2.01).
- Remove the complete wheels ( D2.01).
-  Remove the rear wheels.

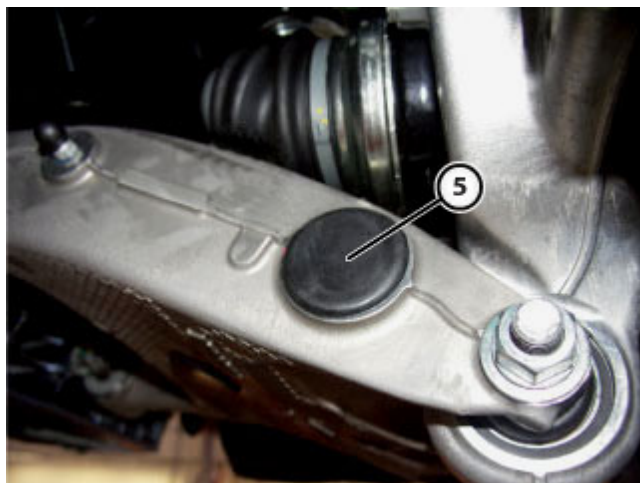


- Disconnect the connector (1), cutting the indicated cable ties.
- Undo the screws indicated.

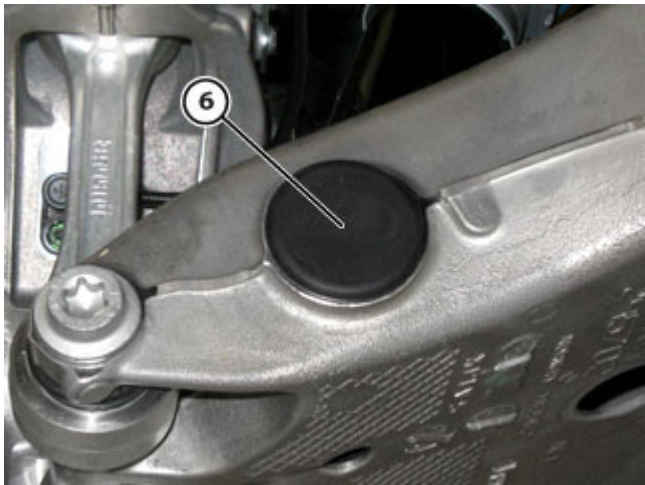


For vehicles with suspension "Lift" system

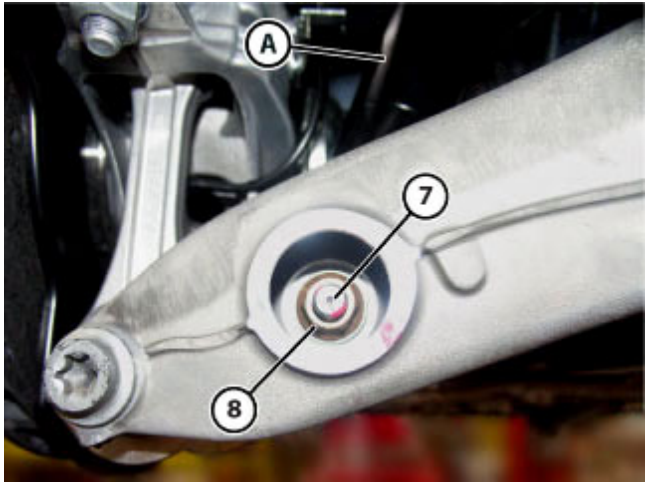
- Undo and disconnect the union (2).
- Remove the clip (3).
- Detach the union (4).



- Prise out and remove the rubber plug (5).



- Prise out and remove the rubber plug (6).



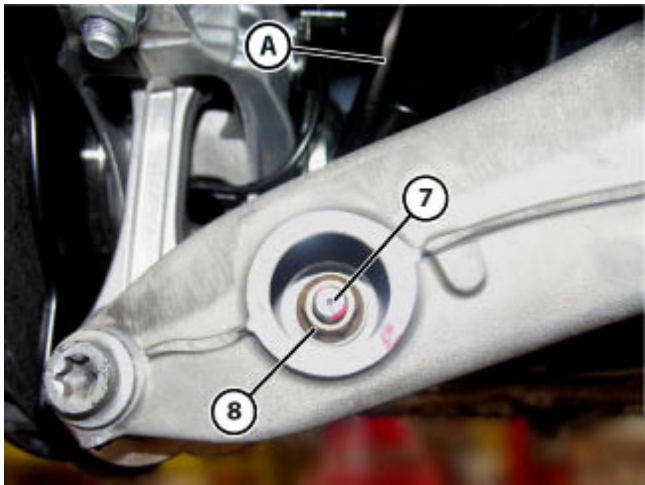
- Undo the nut (8), while preventing the pin (7) from turning.
- Remove the pin (7).

- Remove the rear shock absorber.

Refitting the rear shock absorbers

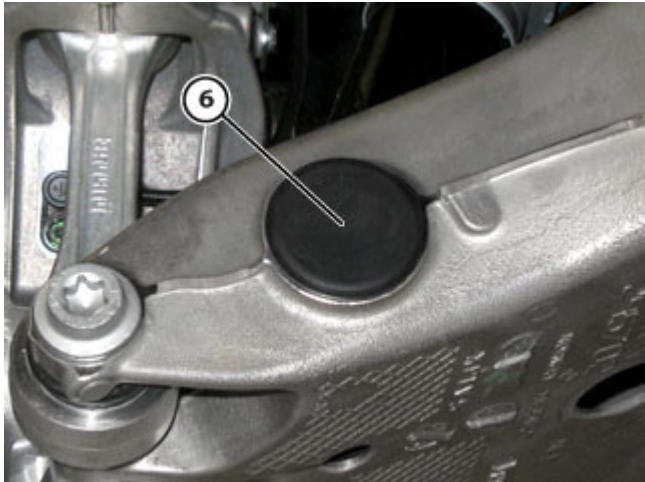
			
Tightening torque		Nm	Class
Fastening the rear shock absorbers	Nut	80 Nm	A
	Screw (pretightening)	20 Nm	A
	Screw	110 Nm	A
Fastening "Lift" system pipe union on front and rear suspension	Union	16 Nm	B

- Fit the rear shock absorber.

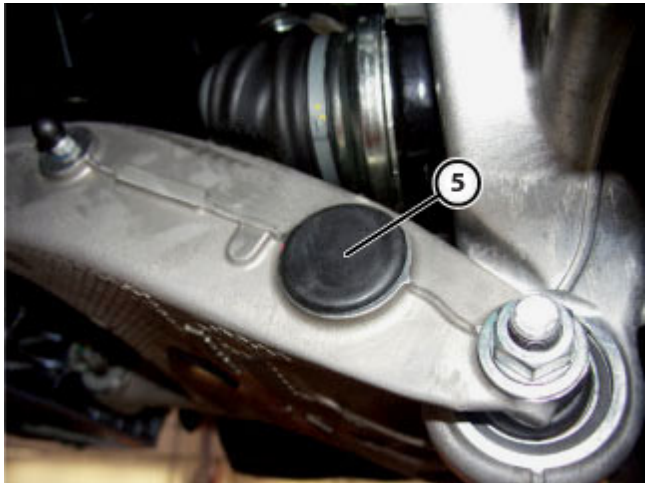


- Install the pin (7).
- Tighten the nut (8), while preventing the pin (7) from turning.

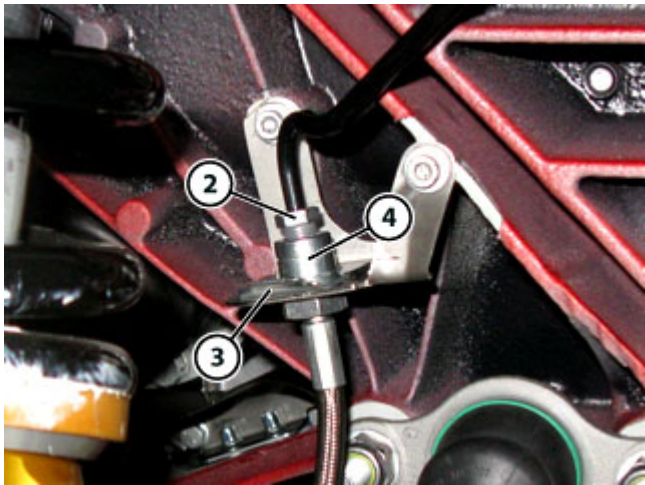
		
Tightening torque	Nm	Class
Nut	80 Nm	A



- Fit the plug (6), pushing into place by hand.



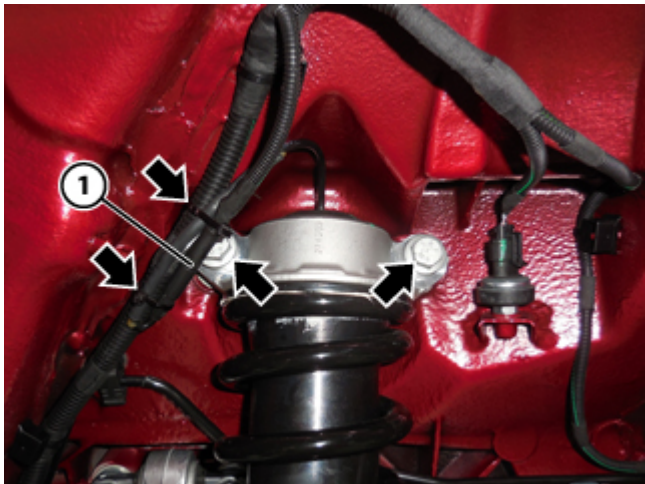
- Fit the plug (5), pushing into place by hand.



For vehicles with suspension "Lift" system

- Fit the union (4).
- Fit the clip (3).
- Connect and tighten the union (2).

		
Tightening torque	Nm	Class
Union	16 Nm	B



- Pre-tighten and definitively tighten the indicated screws.

		
Tightening torque	Nm	Class
Screw (pretightening)	20 Nm	A
Screw	110 Nm	A

- Connect the connector (1), fitting and tightening new clamps as indicated.

- Connect the battery (🔧 F2.01).
- For vehicles with suspension "Lift" system. Fill and bleed the suspension "Lift" system (🔧 D4.08).
- Refit the complete wheels (🔧 D2.01).
 - Refit the rear wheels.
- Adjust the vehicle suspension/wheel alignment (🔧 D4.06).

Replacing the front shock absorber elements

			
Tightening torque		Nm	Class
Fastening bushing and shock absorber stop cap	Nut	40 Nm	A
Fastening connector to shock absorber strut	Nut	40 Nm	A
Fastening front shock absorber link	Screw	90 Nm	A

- Remove the front shock absorbers (🔧 D4.04).

Disassembling variable damping rate magnetorheological shock absorbers

- Undo the screw **(1)**.
- Remove the link **(A)**.



- Immobilise the shock absorber.
- Undo the nut **(3)**.



The following procedure for removing the nut must be observed precisely in order to compress and remove the spring safely.

- Partially loosen the nut **(4)** by just enough to decompress the spring.
- Compress the spring **(B)** with a suitable press.
- Undo the nut **(4)** completely.
- Retrieve the washer **(5)**.
- Slowly release the spring **(B)**.
- Remove the upper stop cap **(C)**, complete with hyperblock.
- Remove the bushing **(6)**.
- Remove the upper spring shim **(D)**.
- Remove the spring **(B)**.
- Remove the bushing **(F)**.
- Remove the lower spring shim **(G)**.

Disassembling variable damping rate magnetorheological shock absorbers for "Lift" system (optional)

- Undo the screw **(1)**.
- Remove the link **(A)**.



- Immobilise the shock absorber.
- Undo the nut **(3)**.



The following procedure for removing the nut must be observed precisely in order to compress and remove the spring safely.

- Partially loosen the nut **(4)** by just enough to decompress the spring.
- Compress the spring **(B)** with a suitable press.
- Undo the nut **(4)** completely.
- Retrieve the washer **(5)**.
- Slowly release the spring **(B)**.
- Remove the upper stop cap **(C)**, complete with hyperblock.
- Remove the bushing **(6)**.
- Remove the upper spring shim **(D)**.
- Remove the spring **(B)**.
- Remove the bushing **(F)**.
- Remove the lower spring shim **(G)**.
- Remove the actuator **(H)**.

Reassembling variable damping rate magnetorheological shock absorbers



- Apply grease **TUTELA ZETA 2** to the surface of the lower shim **(G)** in contact with the spring **(B)**.
- Fit the lower shim **(G)** on the shock absorber.
- Install the bushing **(F)**.
- Fit the spring **(B)**.
- Apply grease **TUTELA ZETA 2** to the surface of the upper shim **(D)** in contact with the spring **(B)**.
- Fit the upper shim **(D)** on the shock absorber.
- Fit the bushing **(6)**.
- Fit the upper stop cap **(C)**, complete with hyperblock.
- Compress the spring **(B)** with a suitable press.
- Fit the washer **(5)**.



Do not release the spring until the nut is fastened to the shock absorber strut.

- Apply thread lock compound **AREXONS SYSTEM 52A43 - MEDIUM STRENGTH THREAD LOCK** to the thread of the nut **(4)** and tighten.

		
Tightening torque	Nm	Class
Nut	40 Nm	A

- Tighten the nut **(3)**.

		
Tightening torque	Nm	Class
Nut	40 Nm	A

- Slowly release the spring **(B)**.



- Fit the link (A).
- Tighten the screw (1).

		
Tightening torque	Nm	Class
Screw	90 Nm	A

Reassembling variable damping rate magnetorheological shock absorbers for "Lift" system (optional)



- Apply grease **TUTELA ZETA 2** to the surface of the shock absorber in contact with the actuator **(H)**.
- Fit the actuator **(H)**.
- With the shock absorber oriented as it would be in the vehicle, the actuator bleed valve **(H)** must face towards the front of the vehicle and be turned by **15° ± 1°** toward the exterior of the vehicle.

- Fit the lower shim **(G)**.
- Install the bushing **(F)**.
- Fit the spring **(B)**.
- Apply grease **TUTELA ZETA 2** to the surface of the upper shim **(D)** in contact with the spring **(B)**.
- Fit the upper shim **(D)**.
- Fit the bushing **(6)**.
- Fit the upper stop cap **(C)**, complete with hyperblock.
- Compress the spring **(B)** with a suitable press.
- Fit the washer **(5)**.



Do not release the spring until the nut is fastened to the shock absorber strut.

- Apply thread lock compound **AREXONS SYSTEM 52A43 - MEDIUM STRENGTH THREAD LOCK** to the thread of the nut **(4)** and tighten.

		
Tightening torque	Nm	Class
Nut	40 Nm	A

- Tighten the nut **(3)**.

		
Tightening torque	Nm	Class
Nut	40 Nm	A

- Slowly release the spring **(B)**.



- Fit the link (A).
- Tighten the screw (1).

		
Tightening torque	Nm	Class
Screw	90 Nm	A

- Adjust the length of the shock absorbers ( D4.04).
- Refit the front shock absorbers ( D4.04).

Replacing the rear absorber elements

			
Tightening torque		Nm	Class
Fastening bushing and shock absorber stop cap	Nut	40 Nm	A
Fastening connector to shock absorber strut	Nut	40 Nm	A

- Remove the rear shock absorbers ( D4.04).

Disassembling variable damping rate magnetorheological shock absorbers



- Immobilise the shock absorber.
- Undo the nut (2).



The following procedure for removing the nut must be observed precisely in order to compress and remove the spring safely.

- Partially loosen the nut (3) by just enough to decompress the spring.
- Compress the spring (A) with a suitable press.
- Undo the nut (3) completely.
- Retrieve the washer (4).
- Slowly release the spring (A).
- Remove the upper stop cap (B), complete with hyperblock.
- Remove the bushing (5).
- Remove the upper spring shim (C).
- Remove the spring (A).
- Remove the plug (D).
- Remove the lower spring shim (E).

Disassembling variable damping rate magnetorheological shock absorbers for "Lift" system (optional)



- Immobilise the shock absorber.
- Undo the nut (2).



The following procedure for removing the nut must be observed precisely in order to compress and remove the spring safely.

- Partially loosen the nut (3) by just enough to decompress the spring.
- Compress the spring (A) with a suitable press.
- Undo the nut (3) completely.
- Retrieve the washer (4).
- Slowly release the spring (A).
- Remove the upper stop cap (B), complete with hyperblock.
- Remove the bushing (5).
- Remove the upper spring shim (C).
- Remove the spring (A).
- Remove the plug (D).
- Remove the lower spring shim (E).
- Remove the actuator (F).

Reassembling variable damping rate magnetorheological shock absorbers



- Apply grease **TUTELA ZETA 2** to the surface of the lower shim **(E)** in contact with the spring **(A)**.
- Fit the lower shim **(E)** on the shock absorber.
- Fit the plug **(D)**.
- Fit the spring **(A)**.
- Apply grease **TUTELA ZETA 2** to the surface of the upper shim **(C)** in contact with the spring **(A)**.
- Fit the upper shim **(C)** on the shock absorber.
- Fit the bushing **(5)**.
- Fit the upper stop cap **(B)**, complete with hyperblock.
- Compress the spring **(A)** with a suitable press.
- Fit the washer **(4)**.



Do not release the spring until the nut is fastened to the shock absorber strut.

- Apply thread lock compound **AREXONS SYSTEM 52A43 - MEDIUM STRENGTH THREAD LOCK** to the thread of the nut **(3)** and tighten.

		
Tightening torque	Nm	Class
Nut	40 Nm	A

- Tighten the nut **(2)**.

		
Tightening torque	Nm	Class
Nut	40 Nm	A

- Fit and connect the connector **(1)**.
- Slowly release the spring **(A)**.

Reassembling variable damping rate magnetorheological shock absorbers for "Lift" system (optional)



- Apply grease **TUTELA ZETA 2** to the surface of the shock absorber in contact with the actuator **(F)**.
- Fit the actuator **(F)**.
- With the shock absorber oriented as it would be in the vehicle, the actuator bleed valve **(F)** must face towards the rear of the vehicle.
- Fit the lower shim **(E)**.
- Apply grease **TUTELA ZETA 2** to the surface of the lower shim **(E)** in contact with the spring **(A)**.
- Fit the plug **(D)**.
- Fit the spring **(A)**.
- Apply grease **TUTELA ZETA 2** to the surface of the upper shim **(C)** in contact with the spring **(A)**.
- Fit the upper shim **(C)**.
- Fit the bushing **(5)**.
- Fit the upper stop cap **(B)**, complete with hyperblock.
- Compress the spring **(A)** with a suitable press.
- Fit the washer **(4)**.

**CAUTION**

Do not release the spring until the nut is fastened to the shock absorber strut.

- Apply thread lock compound **AREXONS SYSTEM 52A43 - MEDIUM STRENGTH THREAD LOCK** to the thread of the nut **(3)** and tighten.

		
Tightening torque	Nm	Class
Nut	40 Nm	A

- Tighten the nut **(2)**.

		
Tightening torque	Nm	Class
Nut	40 Nm	A

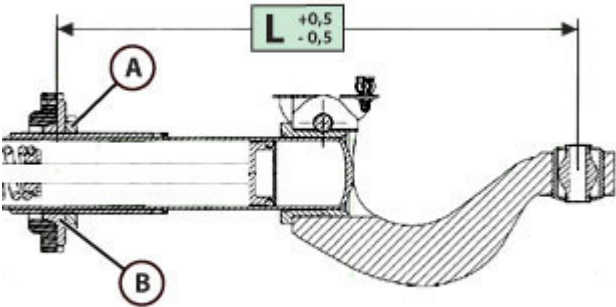
- Slowly release the spring **(A)**.

- Adjust the length of the shock absorbers ( **D4.04**).
- Refit the rear shock absorbers ( **D4.04**).

Adjusting shock absorber length

			
Tightening torque		Nm	Class
Fastening shock absorber stop cap	Ring nut	65 Nm	B

Front shock absorbers



- Loosen the ring nut **(A)**.
 - Tighten or loosen the spring retainer **(B)** to obtain the length **(L)** indicated as follows.
-  The length **(L)** from the spring retainer to the lower fulcrum axis must be adjusted in relation to the type of spring fitted and the type of shock absorber on which the spring is installed. Two different spring sizes are available with different control load tolerances. These are identifiable by a coloured marking stamped on a coil turn.

Length of front variable setting shock absorbers

Yellow	368.5 mm
White	366.5 mm

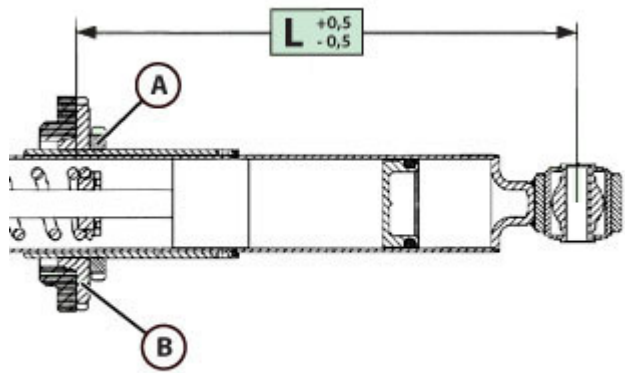
Length of front variable setting shock absorbers (for vehicles with lift system)

Yellow	353.5 mm
White	351.5 mm

- Holding the spring retainer **(B)** to prevent it from turning, tighten the ring nut **(A)**.

		
Tightening torque	Nm	Class
Ring nut	65 Nm	B

Rear shock absorbers



- Loosen the ring nut **(A)**.
- Tighten or loosen the spring retainer **(B)** to obtain the length **(L)** indicated as follows.
- i* The length **(L)** from the spring retainer to the lower fulcrum axis must be adjusted in relation to the type of spring fitted and the type of shock absorber on which the spring is installed.
- i* Two different spring sizes are available with different control load tolerances, identifiable by the different colours stamped on a coil turn.

Length of rear variable setting shock absorbers

Yellow	264.3 mm
White	262.9 mm

Length of rear variable setting shock absorbers (for vehicles with lift system)

Yellow	254.3 mm
White	252.9 mm

- Holding the spring retainer **(B)** to prevent it from turning, tighten the ring nut **(A)**.

		
Tightening torque	Nm	Class
Ring nut	65 Nm	B